18. option(c)

$$Rate = \frac{d[N_2O_5]}{dt} = \left(\frac{1}{2}\right)\frac{d[NO_2]}{dt} = \frac{2\,d[O_2]}{dt}$$

Given that

$$\frac{d[N_2O_5]}{dt} = 6.5 \times 10^{-2}\ \text{mol}\ L^{-1}s^{-1}$$

$$\frac{d[NO_2]}{dt} = 2 \times 6.5 \times 10^{-2} = 1.3 \times 10^{-1}\ \text{mol}\ L^{-1}s^{-1}$$

$$\frac{d[O_2]}{dt} = \frac{6.5 \times 10^{-2}}{2} = 3.25 \times 10^{-2}\ \text{mol}\ L^{-1}s^{-1}$$

19. option(d)

$$H_2O_2 \longrightarrow H_2O + \tfrac{1}{2}O_2$$

$$Rate = \frac{-d[H_2O_2]}{dt} = \frac{d[H_2O]}{dt} = \frac{2d[O_2]}{dt}$$

$$\text{no of moles of oxygen} = \left(\frac{48}{32}\right) = 1.5\ \text{mol}$$

$\therefore$ rate of formation of oxygen $= 1.5\ \text{mol}\ \text{min}^{-1}$

$\therefore$ rate of formation of water $= 2 \times 1.5 = 3\ \text{mol}\ \text{min}^{-1}$

20. option(a)

For a first order reaction $t_{1/2}$ is independent of initial concentration .i.e., $\therefore\ n \neq 1$; for such cases

$$t_{1/2}\ \alpha\ \frac{1}{[A_0]^{n-1}} \quad \text{------- (1)}$$

If $[A_0] = 2[A_0]$; then $t_{1/2} = 2t_{1/2}$

$$2\,t_{1/2}\ \alpha\ \frac{1}{[2\,A_0]^{n-1}} \quad \text{------ (2)}$$

$$\frac{(2)}{(1)} \Rightarrow$$

$$2 = \frac{1}{[2A_0]^{n-1}} \times \frac{[A_0]^{n-1}}{1}$$

$$2 = \frac{[A_0]^{n-1}}{[2A_0]^{n-1}}$$

Answers.indd 248 22-12-2022 19:40:24